Basic Introduction

Accessing Documents

General Information

STREAM

DELTA PRO

DELTA 2

DELTA 2 MAX

Smart Home Panel

Smart Plug

PowerStream Micro-inverter

Wave Air Conditioner

Glacier

Power Kits

PowerOcean

DELTA Pro Ultra

DELTA Pro 3

Smart Home Panel II

River 2 Pro

Code package download

Support

Host

Key	Value
host	https://api-e.ecoflow.com

HTTP access steps

Create a signature with accesskey and secretkey for authentication when HTTP requesting.

How to implement signature:

Step 1: request parameters must be sorted by ASCII value and concatenated with characters =, &

E.g. str=param1=value1&param2=value2

Step 2: if the type is nested, expand and splice according to the method of step 1.

E.g. deviceInfo.id=1&deviceList[0].id=1&deviceList[1].id=2&ids[0]=1&ids[1]=2&ids[2]=3&name=demo

```
{
    // General Parameter, expand to: name=demo1
    "name" : "demo1" ,
    // Array, expand to: ids[0]=1&ids[1]=2&ids[2]=3
    "ids" :[1, 2, 3],
    // Object, expand to: deviceInfo.id=1
    "deviceInfo" :{
        "id" :1
    },
    // Object Array, expand to: deviceList[0].id=1&deviceList[1].id=2
    "deviceList" :[
        {
            "id" :1
        },
        {
            "id" :2
        }
    ]
}
```

Step 3: concatenate accessKey, nonce, timestamp

E.g. str=param1=value1&param2=value2&accessKey=***&nonce=...&timestamp=...

Step 4: encrypt

E.g. byte[] signBytes = HMAC-SHA256(str, secretKey)

Step 5: convert byte[] to hexadecimal string. String sign = bytesToHexString(signBytes)

E.g. sign=85776ede686fe4783eac48135b0b1748ba2d7e9bb7791b826dc942fc29d4ada8

Step 6: add accessKey, nonce, timestamp, sign to HTTP header

E.g.

```
// HTTP Header
accessKey:ba2d7e9...
nonce:123456
timestamp:1671171709428
sign:85776ede686fe...
```

Step 7: get request data according to Content-Type header

If Content-Type is "application/json;charset=UTF-8", get request data from request body, else get from

E.g.

```
// HTTP header content-type=application/json;charset=UTF-8
// HTTP body
{
 "name": "...",
 "begin_time": 123456789
}

// HTTP header content-type is none or other
// HTTP query string
https://${host}//iot-open/sign/device/quota/all?sn=123
```

Step8: verify that the steps to generate the signature are correct

```
accessKey=Fp4SvIprYSDPXtYJidEtUAd1o
secretKey=WIbFEKre0s6sLnh4ei7SPUeYnptHG6V
nonce=345164
timestamp=1671171709428

// HTTP json Body
{
    "sn" : "123456789" ,
    "params" :{
        "cmdSet" :11,
        "id" :24,
        "eps" :0
    }
}

// key-value string
params.cmdSet=11&params.eps=0&params.id=24&sn=123456789&accessKey=Fp4SvIprYSDPXtYJidEtUAd1o&nc

// sign. If the signature you generated is this value, than is correct
07c13b65e037faf3b153d51613638fa80003c4c38d2407379a7f52851af1473e
```

General HTTP Header

Field	Field's Type	Required
accessKey	string	Yes
nonce	int	Yes
timestamp	long	Yes
sign	string	Yes

Query the user's bound device list

Only returns the device bound to itself, not by share.

Key	Value
Method	GET
URL	https://${host}/iot-open/sign/devi
Content-Type	none

Request Body

None

Response Body

Field	Field's Type	Required
code	string	Yes
message	string	
data	List	

DeviceInfo

Field	Field's Type	Required
sn	string	Yes
deviceName	string	Yes
online	int	Yes

Example

```
// request
curl -X GET https://api-e.ecoflow.com/iot-open/sign/device/list \
-H 'accessKey:OCHzRuj6NLF7o43***' \
-H 'timestamp:1681796503289' \
-H 'nonce:234762' \
-H 'sign:f560c3e31d96ad31e4567939f9b3dca7b2c454ca7003f60***'

// response
{
    "code":"0",
    "message":"Success",
    "data":[
        {
            "sn":"DCABZ****",
            "online":1
        }
    ]
}
```

Setting device's function

Key	Value
Method	PUT
URL	https://${host}/iot-open/sign/devi
Content-Type	application/json;charset=UTF-8

Request Body

Field	Field's Type	Required
sn	string	Yes
params	SetCmdRequest	Yes

Response Body

Field	Field's Type	Required
code	string	Yes
message	string	

Example

```
// Delta Pro: Setting the X-Boost switch
// Request
curl -X PUT https://api-e.ecoflow.com/iot-open/sign/device/quota \
-H 'Content-Type:application/json;charset=UTF-8' \
-H 'accessKey:OCHzRuj6NLF7o43***' \
-H 'timestamp:1681872798000' \
-H 'nonce:238752' \
-H 'sign:0f3a1b102cd9a306307b7a9a0f0a9add7b8bfc93cf11***' \
-d '{"sn":"DCABZ***","params":{"cmdSet":32,"id":66,"enabled":1}}'

// Response
{
    "code":"0",
    "message":"Success"
}
```

Query the device's quota infomation

Key	Value
Method	POST
URL	https://${host}/iot-open/sign/devi
Content-Type	application/json;charset=UTF-8

Request Body

Field	Field's Type	Required
sn	string	Yes
params	GetCmdRequest	Yes

Response Body

Field	Field's Type	Required
code	string	Yes
message	string	
data	GetCmdResponse	

Example

```
// Delta Pro: Get the X-Boost switch value
// Request
curl -X POST https://api-e.ecoflow.com/iot-open/sign/device/quota \
-H 'Content-Type:application/json;charset=UTF-8' \
-H 'accessKey:OCHzRuj6NLF7o43***' \
-H 'timestamp:1681872798000' \
-H 'nonce:238752' \
-H 'sign:1e48307441d06a6b31d4c0d66be1580c7748fd0***' \
-d '{"sn":"DCABZ***","params":{"cmdSet":32,"id":66,"quotas":["inv.cfgAcEnabled"]}}'

// Response
{
    "code":"0",
    "message":"Success",
    "data":{
        "inv.cfgAcEnabled":1
    }
}
```

Query device's all quota infomation

Key	Value
Method	GET
URL	https://${host}/iot-open/sign/devi
Content-Type	none

Request Query String

Field	Field's Type	Required
sn	string	Yes

Response Body

Field	Field's Type	Required
code	string	Yes
message	string	

Field	Field's Type	Required
data	GetAllQuotaResponse	

Example

```
// Get Delta Pro all quota values
// Request
curl -X GET https://api-e.ecoflow.com/iot-open/sign/device/quota/all?sn=DCABZ*** \
-H 'accessKey:OCHzRuj6NLF7o43***' \
-H 'timestamp:1681872798000' \
-H 'nonce:238752' \
-H 'sign:534215ccfe93979588975630ea0ad7091c7e250ae5***'

// Response
{
    "code":"0",
    "message":"Success",
    "data":{
        "bmsMaster.soc":"100",
        "bmsMaster.temp":"34",
        "bmsMaster.inputWatts":"0",
        "bmsMaster.outputWatts":"0",
        "pd.remainTime":"14781",
        "inv.cfgAcEnabled":"0",
        "mppt.carState":"0",
        "pd.usb1Watts":"0",
        "pd.usb2Watts":"0",
        "pd.qcUsb1Watts":"0",
        ...
    }
}
```

MQTT certificate acquisition

Get the MQTT certification, using it for MQTT communication.

Key	Value
Method	GET
URL	https://${host}/iot-open/sign/cert
Content-Type	none

Request Body

None

Response Body

Field	Field's Type	Required
code	string	Yes
message	string	
data	AppCertification	

AppCertification

Field	Field's Type	Required
certificateAccount	string	Yes
certificatePassword	string	Yes
url	string	Yes
port	string	Yes
protocol	string	Yes

Example

```
// Request
curl -X GET https://api-e.ecoflow.com/iot-open/sign/certification \
-H 'accessKey:OCHzRuj6NLF7o43***' \
-H 'timestamp:1681872798000' \
-H 'nonce:238752' \
-H 'sign:0396ef9748bf273ecf6ad7ded6333f343d778945f7***'

// Response
{
    "code":"0",
    "message":"Success",
    "data":{
        "certificateAccount":"open-57c134518b5***",
        "certificatePassword":"959253cc103a4008***",
        "url":"mqtt.ecoflow.com",
        "port":"8883",
        "protocol":"mqtts"
    }
}
```

MQTT Topic

Usage of Topic	Topic	From
Report device quota	/open/${certificateAccount}/${sn}/quota	device
Set device function	/open/${certificateAccount}/${sn}/set	app
Reply to the set result	/open/${certificateAccount}/${sn}/set_reply	device
Get some quota	/open/${certificateAccount}/${sn/get	app
Reply to the get result	/open/${certificateAccount}/${sn}/get_reply	device
Report Device status	/open/${certificateAccount}/${sn}/status	device

Description

Field	Get from the HTTP** **AP
certificateAccount	GET: /iot-open/sign/certification
sn	GET: /iot-open/sign/device/list

Request Topic's Common Field

Field	Field Type	De
id	string	Messa timest the sa
version	string	Protoc
operateType	string	Device device
from	string	Reque
params	ParamInfo	

Example

```
{
  "id":"123456789",
  "version":"1.0",
  "operateType":"TCP",
  "from":"iOS",
  "params": {
    "cmdSet": 32,
    "id": 33,
    "standByMode": 0
  }
}
```

3.Reply Topic's Common Field

Field	Field Type	De
id	string	Messa timest the sa
version	string	Protoc
operateType	string	Device device

Field	Field Type	De
code	string	Respo belonc
message	string	Error ir
data	DataInfo	Respo

Example

```
{
    "id":123456789,
    "version":"1.0",
    "operateType": "TCP",
    "code": "0",
    "data": {
        "cmdSet": 32,
        "id": 33,
        "ack": 0
    }
}
```

Report Device status

Usage of Topic	Topic	From
Report Device status	/open/${certificateAccount}/${sn}/status	device

Message Format

```
{
  "id":"123456789",
  "version":"1.0",
  "timestamp": 1634841971000,
  "params": {
    "status": 0
  }
}
```

Field

Field	Field Type	De
status	int	Device

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